

Visualization of Data

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Outline

Types of Data

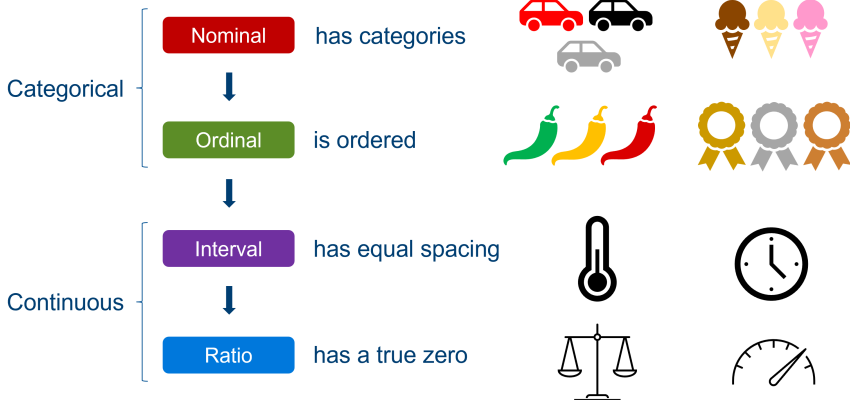
Bar Plot

Histogram

Scatter Plot

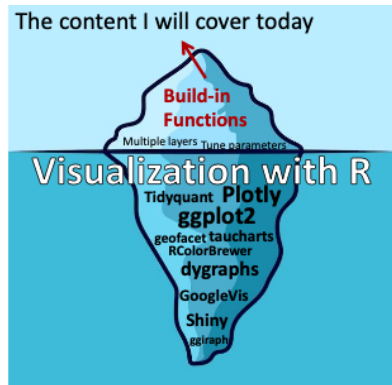
Lab Report

Types of Data



Types of Data Visualization

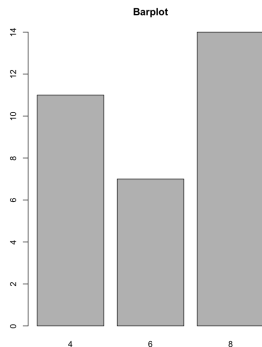
- ▶ Categorical Variable(s):
 - ▶ Bar plot
- ▶ Continuous Variable(s):
 - ▶ Histogram
 - ▶ Scatter plot



Bar Plot

- ▶ Bar plots are used to compare the frequency of different categories.
- ▶ The height of each bar corresponds to the frequency.
- ▶ `barplot()`

```
barplot(data,  
        main = "Title of figure",  
        xlab = "Name of X-axis",  
        ylab = "Name of Y-axis",  
        col = "Color within Bin",  
        border = "Color in border  
                of Bin",  
        xlim = c(a, b), # Modify  
                        range of X-axis  
        ylim = c(a, b) # Modify  
                        range of Y-axis  
        )
```

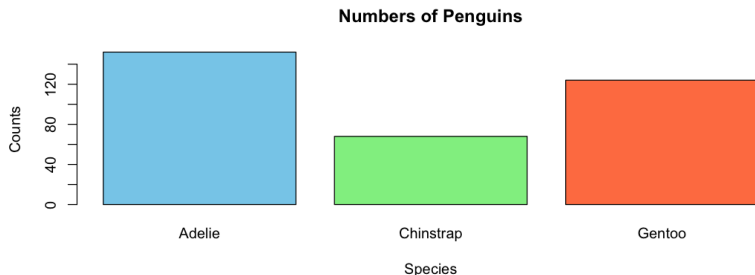


Bar Plot

We'll use the built-in dataset "penguins" to demonstrate

```
species <- table(penguins$species)
barplot(species,
        main = "Numbers of Penguins",
        xlab = "Species",
        ylab = "Counts",
        col = c("skyblue", "orange", "coral"))
```

Check [Colors in R](#) to discover more colors!

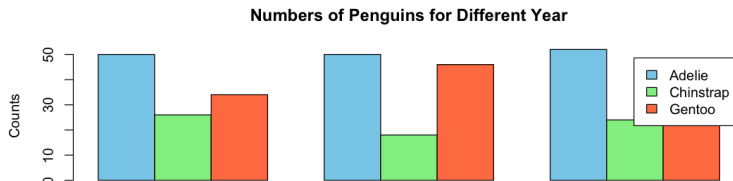


Multiple Levels Bar Plot

We'll keep using the "penguins" data to demonstrate

```
species_year <- table(penguins$species, penguins$
  year)
species_year

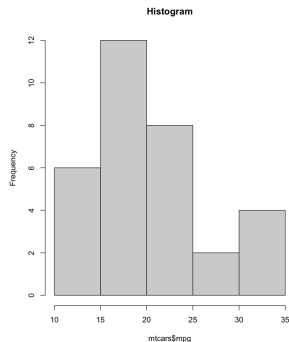
barplot(species_year,
  main = "Numbers of Penguins for Different
    Year",
  xlab = "Year",
  ylab = "Counts",
  beside = TRUE,
  legend = TRUE,
  col = c("skyblue", "lightgreen", "coral"))
```



Histogram

- ▶ Histograms are used to visualize the distribution of continuous variables.
- ▶ The height of each bin corresponds to the frequency.
- ▶ `hist()`

```
hist(data,  
      main = "Title of figure",  
      xlab = "Name of X-axis",  
      ylab = "Name of Y-axis",  
      col = "Color within Bin",  
      border = "Color in border of  
              Bin",  
      breaks = # Specify how many  
               bins we want  
)
```



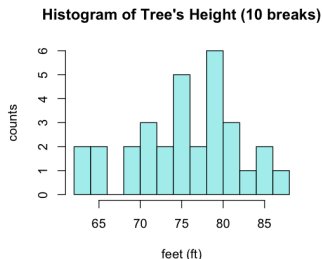
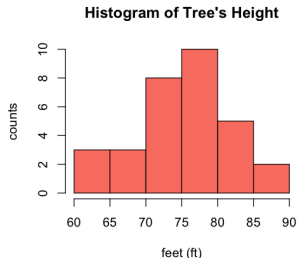
Histogram

We'll use the built-in dataset "trees" to demonstrate

```
hist(trees$Height,  
     main = "Histogram of Tree  
           's Height",  
     xlab = "feet (ft)",  
     ylab = "counts",  
     col = "salmon")
```

```
# Use "break" to set the  
  number of bins
```

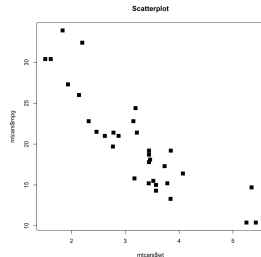
```
hist(trees$Height,  
     main = "Histogram of Tree  
           's Height (10 breaks)",  
     ,  
     xlab = "feet (ft)",  
     ylab = "counts",  
     breaks = 10,  
     col = "paleturquoise")
```



Scatter Plot

- ▶ Scatter plots are used to visualize the relationship between two variables.
- ▶ Each point represents a pair of values.
- ▶ `plot()`

```
plot(data,  
      main = "Title of figure",  
      xlab = "Name of X-axis",  
      ylab = "Name of Y-axis",  
      col = "Color of dot",  
      pch = "Shape of dot",  
      lty = "Shape of line",  
      type = "Type of scatter plot"  
)
```



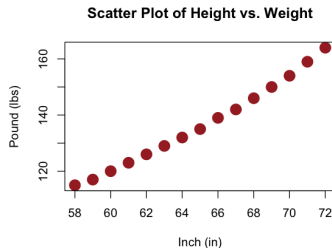
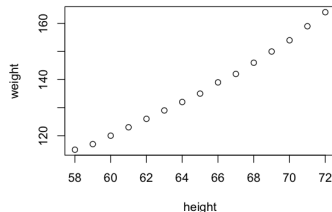
Scatter Plot

We'll use the built-in dataset "women" to demonstrate

```
?women

plot(women)

# Add some settings
plot(women,
      main = "Scatter Plot of
              Height vs. Weight",
      xlab = "Inch (in)",
      ylab = "Pound (lbs)",
      pch = 20,
      col = "brown",
      cex = 3,
      type = "p")
```



Plot Style

Check these websites for more plot styles

- ▶ Colors (col=): <https://r-charts.com/colors/>
- ▶ Plot Symbols (pch=):
<https://r-charts.com/base-r/pch-symbols/>
- ▶ Line Style(lty=) and Plot Type (type=):
<https://r-charts.com/base-r/line-types/>

The End of the Course

1. You don't need to memorize everything!
Google, Stack Overflow, GenAI would be your good friends!
2. Future exploration
 - ▶ Various graph types (<https://r-graph-gallery.com/>)
 - ▶ Plot matrix (combining multiple figures)
 - ▶ tidyverse, ggplot2 (widely used visualization packages)

Lab Report

- ▶ The report file will be in the NTU COOL Assignment section
- ▶ Submit your file in pdf format
- ▶ Deadline: 3/29 (Sun) 23:59
- ▶ Late submission is not accepted